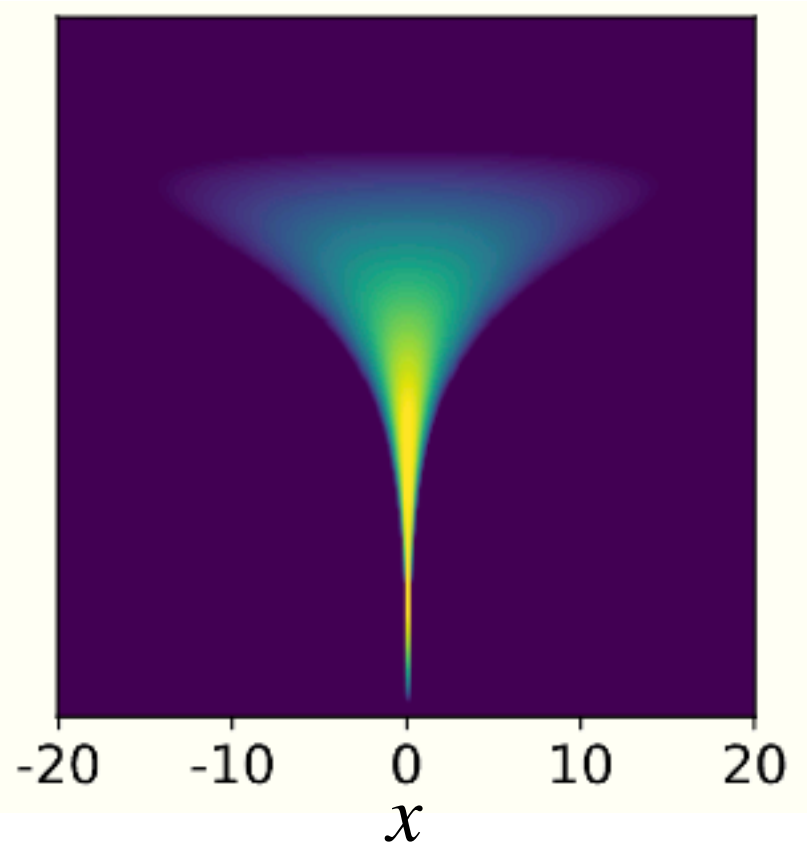
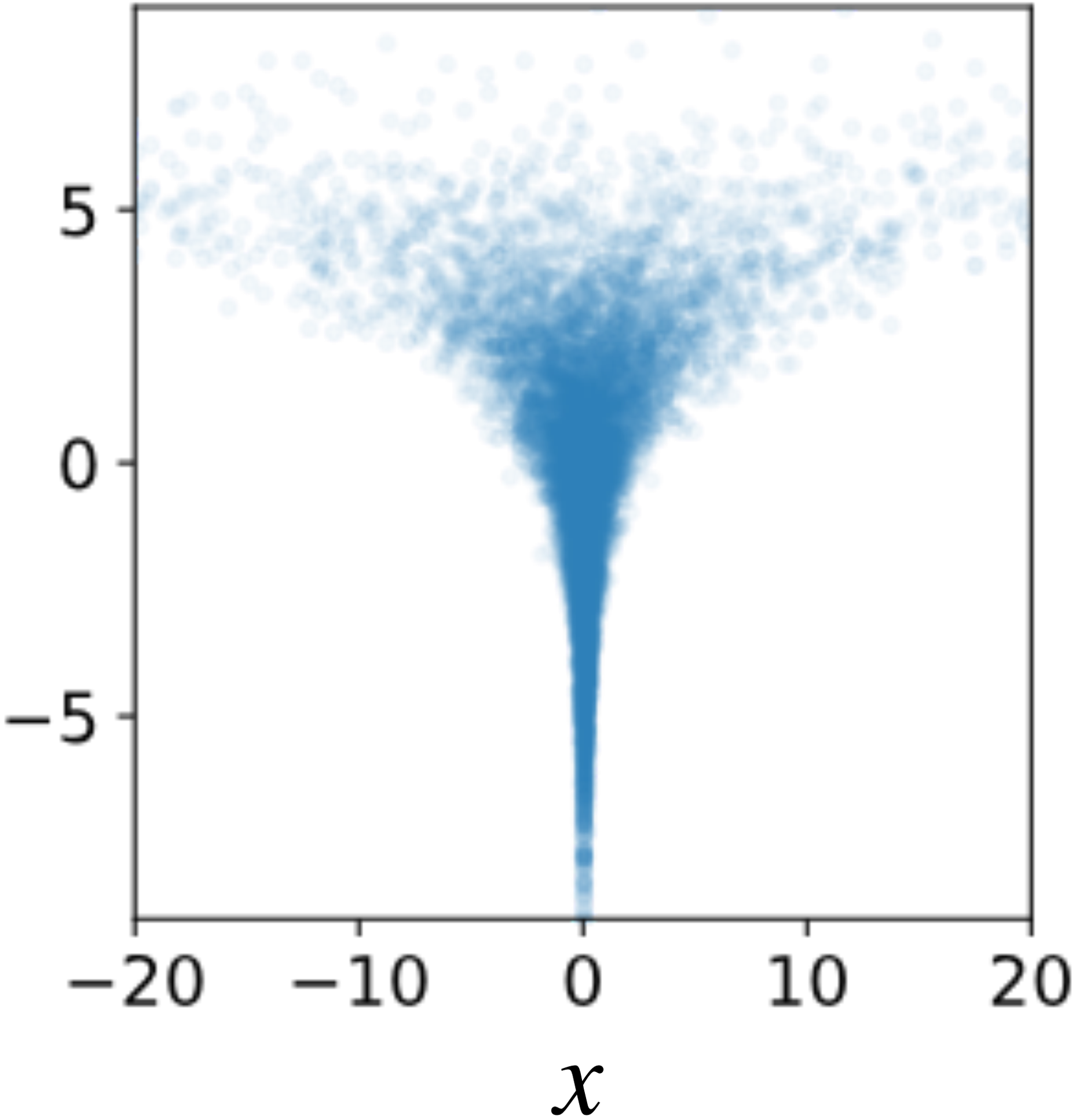
TUNING STEP SIZE

24

- **Example:** Neil's funnel

$$Y \sim \mathcal{N}(0,3), \qquad X \sim \mathcal{N}(0,e^Y)$$




5
0
−5
−20
−10
0
10
20
x

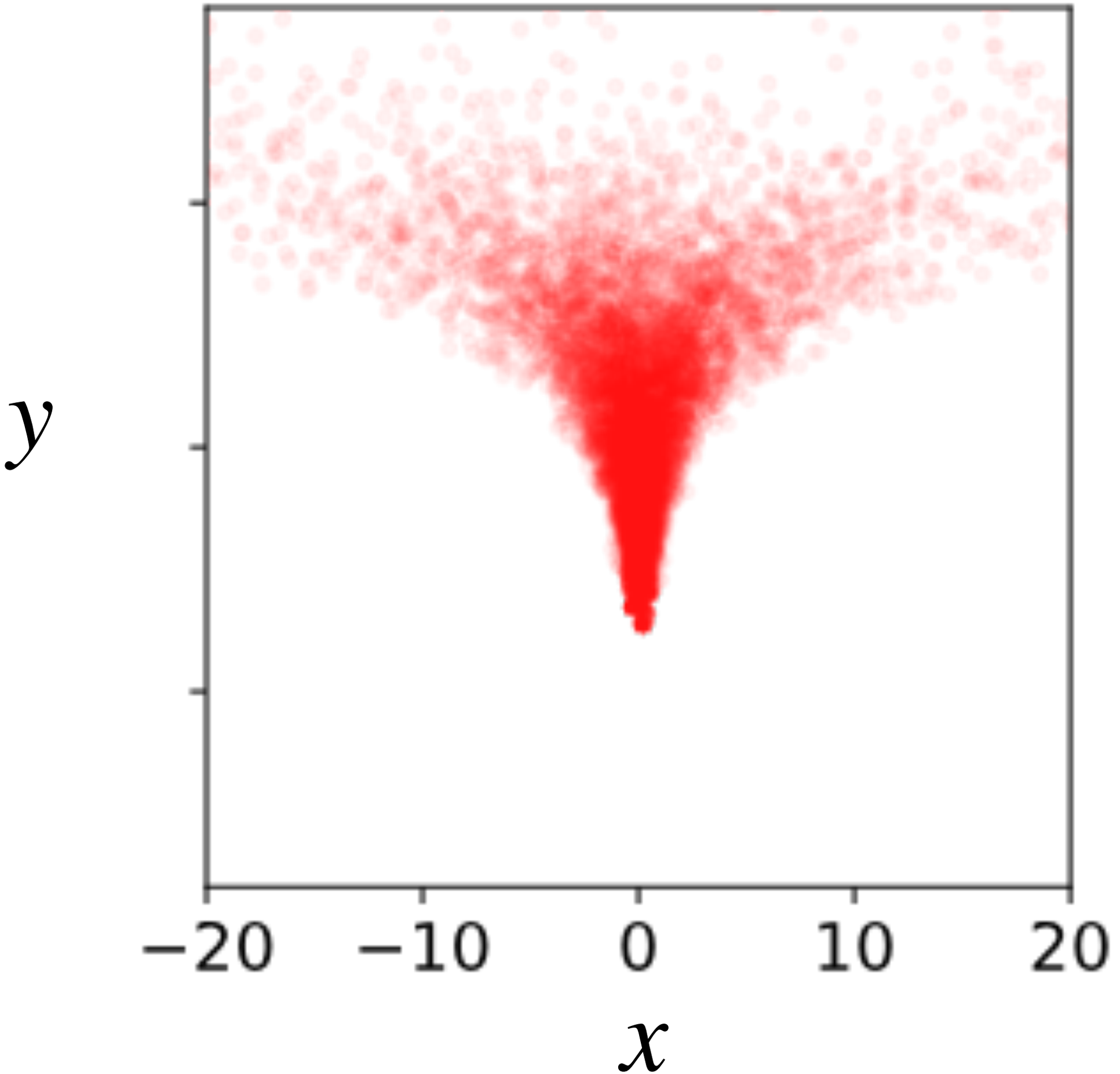
- Step size ϵ is chosen to account for the scale of the target

- When distributionis wide, want to pick a big step size

- When the distribution is narrow want to pick a small step size

- Small ϵ leads to high acceptance rates but small movement

- Large ϵ leads to low acceptance rates and propose samples in unlikely regions


y
−20
−10
0
10
20
x

TUNING STEP SIZE

- Step size ϵ is chosen to account for the scale of the target
  - When distributionis wide, want to pick a big step size
  - When the distribution is narrow want to pick a small step size
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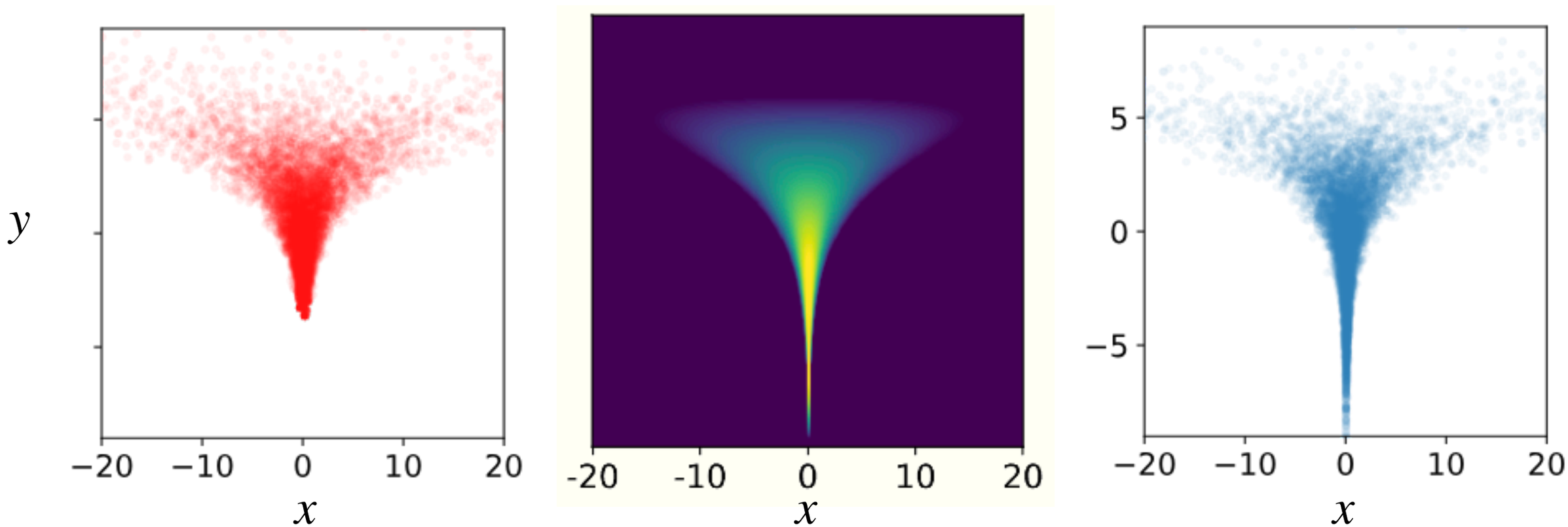

OPTIMAL SCALING